

TrES-3b

R-1962

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140320 · created 2026-08-06T23:01:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456736.8948 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.023
Transit depth (Rp/Rs)^2	2.72 +/- 0.77 [%]
Orbital Inclination (inc)	82.1 +/- 0.72
Airmass coefficient 1 (a1)	1.023 +/- 0.015
Airmass coefficient 2 (a2)	-0.0138 +/- 0.0044
Scatter in the residuals of the lightcurve fit is	1.49 %
Best Comparison Star	#3 - [26, 296]
Optimal Aperture	3.01
Optimal Annulus	6.64
Transit Duration (day)	0.057 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.023 vs 0.16309 (deviation 0.07σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.04σ)
Mid-time O–C	2.0 min
Depth SNR	6.4
Residual scatter	1.49 %

Light curve

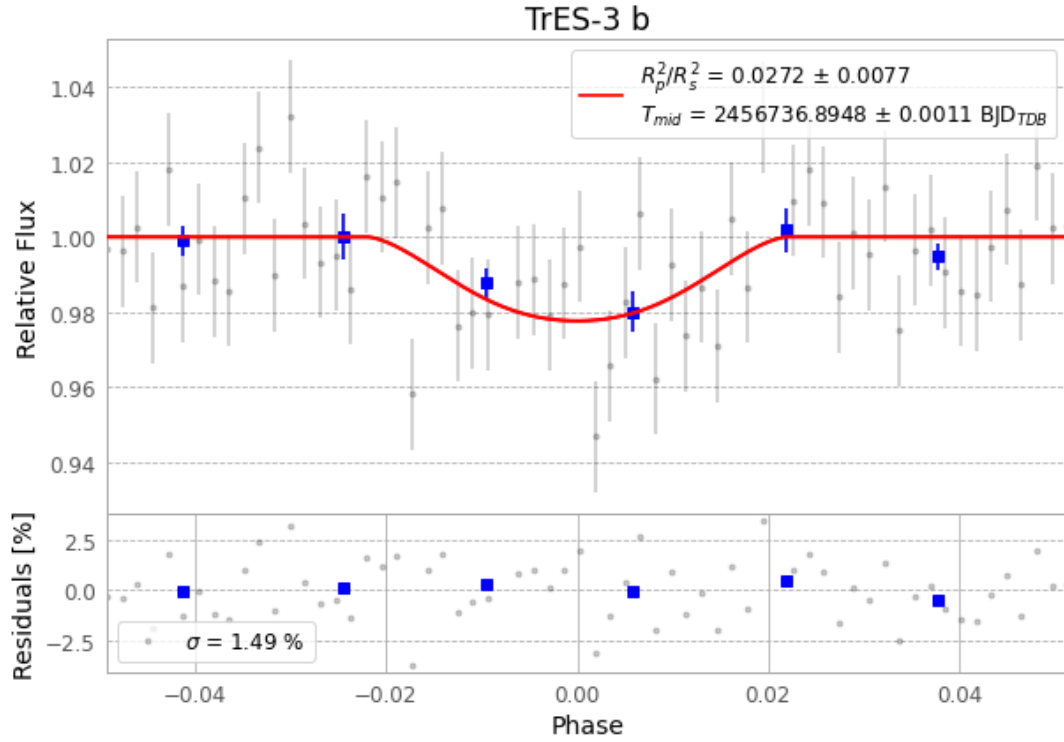


Figure 1 — FinalLightCurve_TrES-3 b_2014-03-20.png (SHA-256 2a42b0715aee7fdf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [224, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ed6097c2610ce14c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), TRES-3140320075412.FITS → TRES-3140320110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140320.log (f95443fe3bba...), FinalLightCurve_TrES-3 b_2014-03-20.png (2a42b0715aee...), FinalLightCurve_TrES-3 b_2014-03-20.csv (a3e6c08f1074...)

Compute resources

Wall time 117.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:01Z.